Making Computer Music Meaningful In Schools

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Abstract

This paper examines the variety of modes of interaction with computers that occur within a school context and relates these to three locations of meaning: the personal, social and cultural. The computer is examined as a tool for teaching and developing teaching materials, as an instrument and as a cognitive amplifier. These ideas are examined through a philosophical argument and supported by vignettes drawn from a participant observation case study that focused upon the meaning of music to students in a school and the processes which give access to that meaning. The central argument revolves around the idea that music making in schools can be 'cutting edge' creative works rather than merely 'school music' and push the boundaries of aesthetic product. When computers are used in these kinds of ways in school music previously abstract concepts and the opportunity to produce high quality music become accessible to children. Finally, the paper argues that through experiencing making music in these highly 'technologised' ways we come to gain an understanding and be critical of how art is made in our every day lives. This serves as a demystification of the processes of music making within the media that pervades our lives.

Making computer music meaningful in schools.

Music is no stranger to technology. The piano key, the trumpet valve and indeed the system of western music notation are all brilliant technologies that had a profound effect upon how humanity expressed itself through music. Each of those technologies was seized upon by composers and performers alike to increase the range and expressiveness of the music they made. In this paper, I propose that the so-called new technologies of the synthesizer and the computer are new surfaces in which we inscribe meaning. Through examining a philosophical shift in the way, we view the modes of engaging with technology in music education, I hope to argue that music making in schools can become more relevant meaningful and 'cutting-edge'. To illustrate the practical application of the principles discussed here I will also draw narrative vignettes taken from a recent long-term participant-observation cases study (Dillon, 2000).

The philosophical argument.

Technology is often described as a tool. Brown (1999) suggests that this is a limiting metaphor. In education, this is often the case. Teachers who are often unfamiliar users of technology use the word in this manner. Those in technology related studies also perceive the human machine interaction as a user and tool metaphor. Historically, artists have a different kind of relationship with technology. This kind of creative rather than reactionary relationship is of interest here. We do not applaud the piano-key mechanism for its technical beauty. Although in engineering terms it may have been a past century equivalent to a major increase in cheap computer memory, or speed. Rather this innovation increased attention to the dynamic and expressive qualities of music of the Romantic period. Likewise, the trumpet valve increased the range and flexibility of brass instruments. The response from composers was to compose music that stretched the aesthetic

boundaries of those instruments and of music itself. The important idea here is the focus upon stretching aesthetic boundaries rather than the technical brilliance of the tool. Brown (1997) suggests that we perceive technology in multiple ways- not merely as a tool. Amongst the modes of engagement he suggests are computer/ synthesizer as instrument, as cognitive amplifier (Papert, 1980), and as a tool.

What these ideas imply are that the technology takes on different roles and different modes of engagement depending upon our needs and uses for them. In musical terms the implications are quite far reaching. Firstly, to consider the computer and the synthesizer as an instrument requires the traditional notion of a developmental pedagogy and a repertoire that stretches and expands the expressive qualities of that instrument. The notion of cognitive amplifier or assistant suggests that the technology has input to the process of making music that reaches beyond that of a passive tool. It functions as a way of thinking that is external to the mind and body, a kind of assistant in the process. The traditional aspect of tool is still apparent – it does a job that it is designed to do functionally and that is proportional to the effort we afford it. A hammer and nail analogy, the technology has strength where we physically may not. It enables quite specific functions. When we consider the notion of technology having these kinds of characteristics and our modes of engagement with it being varied then we can see that the tool metaphor alone is inadequate for artists and indeed educators. Furthermore, we can see limitations of our preparation for engagement with technology- skills and in the *repertoire* and methods of learning about this-*Education.*

Cage- limitations of notation as opposed to magnetic tape and digital.

John Cages suggests that he '*finds it quite curious that, given such wonderful technological means for creativity, our schools continue to fix their minds on historical means of creating music.* (Reimer & Wright (1992:48). He is referring to the technology of written notation and traditional western art music forms of composition that focus upon this technology for creative music making. In reference to Western harmony he suggests that '*the ideas of harmony and its reference to particular steps in the field of sound,* [as].. *inadequate for the composer, who will be faced with the entire field of sound*. (Cage as cited in Reimer & Wright, 1992:46).He calls these '*limiting sound producing mechanisms, that do not allow for the vast possibilities of such materials as magnetic tape and other contemporary devices for sound production*. (Cage as cited in Reimer & Wright (1992:44). In the study that these statements appear, and are discussed, they are considered radical and perhaps the most opposite to the existing viewpoints of music educators, philosophers and composers. The synthesizer and the computer have had an indisputably large influence on music made in the last 30 years. If we consider the notion of technology as tool being a limiting metaphor then these views become even more clearly relevant to contemporary music making and education. As the body of materials- *repertoire*, produced by these means increases in volume and significance our need to understand, access and develop music in these ways also becomes more necessary.

Where is technology now?

The eminent Australian saxophonist Dr Peter Clinch once compared the synthesizer's position to that of the saxophone at the beginning of the century. Highly used as an instrument in popular music but little organised repertoire and pedagogical development. An instrument played by clarinet players and one that was often excluded from large-scale art ensembles. Even now, this seems to be an

accurate description of the computer and synthesizer position. The limitations of traditional sound production and notation systems are also apparent as the cognitive amplifiers of the last centuries. Even as tools for making music including traditional notation, the digital medium is faster, clearer and has greater capacity for assisting and giving more immediate feedback to the composer. The digital medium has made the once abstract concepts of harmony and sound production more ready to hand, immediate and available. Many concepts of music making that were reserved for more experienced users of the technology of notation are now available to children in accessible form.

Meaning and music making

In my own research into the meaning of music to students in a school context (Dillon, 2000), I have suggested that music has three primary meanings. Firstly, a personal meaning that refers to the intrinsic meaning that is experienced when we make music through personal engagement. The flow of personal experience (Csikszentmihalyi, 1994). Secondly that of social-meaning that we get making music with others in ensembles. Something described by musicians in the study as a wordless meaning, a shared communication (Dillon, 2000). Finally, cultural meaning- when the music that is made and the music maker is affirmed and valued as an artist and for their art by the community. This is a reciprocal interaction between culture and artist. Both the artist and product are contribute to the distinctive character of its 'culture' and are projected by the community as such. I suggestion of this study that meaning in music experience is gained from engagement with all of these interactions. For music to be meaningful students need access to music making experiences that afford the opportunity to experience the full range of meaning from personal through social to cultural.

Implications for practice

What are the implications for practice within a school context?

Firstly, the computer and synthesizer need to be considered instruments in their own right. The AMEB has recently included a pedagogy for 'keyboard' referring to electronic keyboard in its examination structure and in Victoria the Synthesizer is included as an instrument for Solo performance in the Year twelve examination syllabus. This is some evidence of the inclusion of the synthesizer as an instrument; the acceptance of the computer as instrument is a more difficult instance although QUT in its New Media degree includes it as a principal instrument. There is also a need to assemble and commission repertoire or its 'sound project' equivalent so that students of the instrument can learn progressively through repeatedly deeper engagement with the expressiveness of the instrument.

Secondly, The role of computer/ synthesizer as cognitive amplifier for composition arrangement and realisation of musical ideas needs to become part of the process of music making. There needs to be recognition initially of the 'concretizing' aspects of the media that allows it to make previously abstract concepts of large-scale orchestration and arrangement accessible to children. Beyond this, the notion of using these media to create new music rather than simply recreate what Cage call historical music, become increasingly possible. Such ideas as aleatoric music, Algorithmic composition, Granular synthesis, Minimalism and a myriad of even newer forms are available to young composers through digital medium.

Thirdly, the use of the computer as a tool for doing rapid or complex computations frees the composer to experiment with ideas and receive immediate aural and visual

feedback. The composer can focus upon expression rather than technical manipulation.

In relation to meaningful experience, these ideas suggest that students need access to all of these modes of engagement and within the framework of personal, social and cultural experiences. It means that technology must be available to be engaged with in all of these ways. There needs to be the opportunity to make and perform music using this technology and to develop its expressive capabilities to the same kinds of levels as any instrument. It means students need access to e ensemble experiences – synthesizer ensembles, electro acoustic ensembles, sound creation projects and installations and exhibitions of sonic realisations . In a cultural sense, the school community needs to value the person and the work that is made through these kinds of interactions. It needs to do this in pragmatic ways- using music of this kind to project something about the school community. Furthermore, it must also explore purely aesthetic ways, where the instrument and works are given the opportunity to push the boundaries of aesthetic development in the domain.

Case study example

In 1995, I had the opportunity to begin observing how these philosophical ideas worked in practice. I began a participant-observation case study at an independent pre –preparatory to year-twelve school in the eastern suburbs of Melbourne. The following is a description of how I observed those philosophical aspects were addressed in curriculum, co curriculum and as part of the school community.

As an Instrument

The school hired keyboard teachers that were proficient synthesizer players and sound programmers. The instrumental program offered private lessons on the instrument and created a student-centred pedagogy for development. Each synthesizer was required to play in a synthesizer ensemble and a number of electro acoustic ensembles/bands, projects. The instrument was offered with the year-eight classroom-instrumental program and used in classroom music ensembles from grades four to twelve.

As a cognitive amplifier

Within the instrumental program, composing using computer was offered as a private study in the same way as an instrument. Within the classroom program year ten featured synthesizer and computer use within the 20th century composition elective and the computers/ synthesizers were used throughout the classroom program (P-12) as a mode of composing and organising musical materials. There was a conscious effort made for the technology to be ‘transparent’, that is, not used merely for its ‘new’ attractiveness but for its functionality and ready access to experience.

As a tool

Digital software and hardware in the music department was used to make resource materials by teachers and students, publish print music and to give repetitive- drill and response training in aural skills and music notation theory. Not to mention the use of the digital technology to demonstrate and make concrete abstract concepts of harmony and musical structure.

Access to meaning

In relation to access to meaning, students had access to personal engagement though classroom use and private tuition. Students had access to social interaction, through classroom ensembles and school ensembles. Cultural interaction through electronic music concerts and multi media church

services and ceremonial events. These events featured music composed and arranged by teachers and students. Those compositions were commissioned with a brief to serve the pragmatic function of the ceremony and to push the boundaries of the art form. Because of the lack of repertoire for the instruments, they were forced to arrange and compose music of their own. This made the process even more meaningful as the students were affirmed by the school community for their unique contribution to school ceremony and by the domain for their contribution to the development of the idiom. This idea is amplified by the school minister who describes the role of a synthesizer ensemble at a school ceremony:

> What it said to the students was that there was something almost tribal about this. That they could create their own rhythm, their own message, their own particular way of being musical , it wasn't imposed from anywhere else it actually happened through the creativity of the staff at the time and the students who worked with those people. I thought it was an extraordinary time. We all commented later how passionate it was and how people if they didn't remember anything else about that night, remembered that music. (Minister)

This is a powerful affirmation of the meaningfulness of this kind of music making within a school community. Remember that we are not describing students re creating a Beethoven quartet, but students work. The comment is not about a fine performance but a cutting edge, relevant and expressive one.

Discussion of implications

What I am arguing here is that we need to move beyond the limitations of 'technology as tool'. We need to view the modes of engagement with technology as multiple. Then we can both develop the boundaries of creative music making and enhance our access to meaningful and expressive music making experiences. This means that within a school context, music making can be truly creative not merely re-creative. The computer and the synthesizer have the capacity to give access to experiences in music making that previously were only available to an elite few who could navigate the complex representation system/technology. Cage argues that this is a kind of democratisation of the process of making music, where conductor, notation and even performers are no longer necessary (Reimer & Wright, 1992:49). This is reflected strongly in the musical community in popular music, music from a variety of non-western cultures and in contemporary art music. This new access and democratisation makes it possible for cutting-edge music to be made by young composers in a school. For music as it is found in the wider community to be simulated in a school, demystifying the process of music making in media (Schusterman, 1992) and indeed music making per se. It gives access to understanding and experiencing complex musical ideas first hand. To achieve this the first step is to consider a shift to a philosophy that expands the view of technology and human interaction in music making. Next, there is a need to respond to this shift in thinking in the simple ways described in the case study example.

Concluding remarks

So how does this approach make computer music meaningful? Members of the electronic music ensemble at the school suggested that they felt that as a group they were on the 'cutting edge of electronic chamber music'. They attributed this feeling to the sense of uniqueness, control over repertoire development and function in the community. This had a profound effect on the confidence and self-esteem of the participants. The school minister described it as a sense of belonging, '*which has had the effect of raising their own awareness of who they are.'* The students reported

experiencing this sense of belonging on two fronts, firstly the deep relationships developed through playing together in an ensemble and secondly, the relationship with the community gained through interaction and purpose. The dual roles of performing music for purely aesthetic and pragmatic purpose had an effect upon their personal feelings about community and the feelings of the community about them. What I am arguing here is that if we open the school curriculum to view our modes of engagement with technology in music education as multiple as suggested by this paper, then we are making way for more meaningful relationship with expressive music making and surely this is the goal of music education.

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